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Quantum Algorithms HW3

① Proving $(H \otimes H) \text{CNOT}_{10} (H \otimes H) = \text{CNOT}_{01}$ $H|y\rangle = \frac{1}{\sqrt{2}} \sum (-1)^{xy} |x\rangle$

First Hadamard layer $\mapsto (H \otimes H) |x_1, x_0\rangle = \frac{1}{2} \sum (-1)^{x_1 a + x_0 b} |a, b\rangle$

$\hookrightarrow \text{CNOT} \mapsto \text{CNOT}_{10} \left(\frac{1}{2} \sum_{a,b \in \{0,1\}} (-1)^{x_1 a + x_0 b} |a, b\rangle \right) = \dots$
reindexing sum w/ $c = a \oplus b \Rightarrow b = a \oplus c$

$\downarrow$

$$= \frac{1}{2} \sum_{a,c \in \{0,1\}} (-1)^{x_1 a + x_0 (a \oplus c)} |a, c\rangle$$

$$= \frac{1}{2} \sum_{a,c \in \{0,1\}} (-1)^{(x_1 \oplus x_0) a + x_0 c} |a, c\rangle$$

Second Hadamard layer $\mapsto \text{LHS } |x_1, x_0\rangle = \frac{1}{4} \sum (-1)^{(x_1 \oplus x_0) a + x_0 c + a y_1 + c y_0} |y_1, y_0\rangle$

$$= \frac{1}{4} \sum_{y_1, y_0 \in \{0,1\}} \left(\sum_{a \in \{0,1\}} (-1)^{a(x_1 \oplus x_0 \oplus y_1)} \right) \left(\sum_{c \in \{0,1\}} (-1)^{c(x_0 \oplus y_0)} \right) |y_1, y_0\rangle$$

use orthogonality property $\sum_{k \in \{0,1\}} (-1)^{km} = 2\delta_{m,0}$

$$\sum_{a \in \{0,1\}} (-1)^{a(x_1 \oplus x_0 \oplus y_1)} = 2\delta_{y_1, x_1 \oplus x_0}$$

$$\sum_{c \in \{0,1\}} (-1)^{c(x_0 \oplus y_0)} = 2\delta_{y_0, x_0}$$

subbing back ...

$$\text{LHS } |x_1, x_0\rangle = \frac{1}{4} \cdot (2\delta_{y_1, x_1 \oplus x_0}) \cdot (2\delta_{y_0, x_0}) |y_1, y_0\rangle$$

$$= |x_1 \oplus x_0, x_0\rangle$$

$$\text{RHS } |x_1, x_0\rangle = \text{CNOT}_{01} |x_1, x_0\rangle = |x_1 \oplus x_0, x_0\rangle$$

Verify each states

$|00\rangle$:

$$(H \otimes H)|00\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \xrightarrow{CNOT_{01}} \frac{1}{2}(|00\rangle + |01\rangle + |11\rangle + |10\rangle)$$

$|01\rangle$:

$$(H \otimes H)|01\rangle = \frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) \xrightarrow{CNOT_{01}} \frac{1}{2}(|00\rangle - |01\rangle + |11\rangle - |10\rangle)$$

$|10\rangle$:

$$(H \otimes H)|10\rangle = \frac{1}{2}(|00\rangle + |01\rangle - |10\rangle - |11\rangle) \xrightarrow{CNOT_{01}} \frac{1}{2}(|00\rangle + |01\rangle - |11\rangle - |10\rangle)$$

$|11\rangle$:

$$(H \otimes H)|11\rangle = \frac{1}{2}(|00\rangle - |01\rangle - |10\rangle + |11\rangle) \xrightarrow{CNOT_{01}} \frac{1}{2}(|00\rangle - |01\rangle - |11\rangle + |10\rangle)$$

$$b) \quad H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad H \otimes H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

$$\text{where } CNOT_{10} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad CNOT_{01} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$A = CNOT_{10}(H \otimes H)$$

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$$

Multiply $(H \otimes H)A$

$$\text{LHS} = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \\ 0 & 0 & 4 & 0 \\ 0 & 4 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = CNOT_{01}$$

$$\text{Thus, } \underline{(H \otimes H)CNOT_{10}(H \otimes H) = CNOT_{01} \quad \square}$$

②

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$$\text{CNOT}_{01} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$\text{CNOT}_{10} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\text{CNOT}_{10} \cdot \text{CNOT}_{01} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

~~$(\text{CNOT}_{10} \cdot \text{CNOT}_{01}) = \text{CNOT}_{01}$~~

$$\text{CNOT}_{01} \cdot (\text{CNOT}_{10} \cdot \text{CNOT}_{01}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \text{ or SWAP } \square$$